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Frequency of mutations associated with Fluoroquinolone resistance among *Mycobacterium tuberculosis* and their correlation with treatment outcome

Abstract:

Background & objectives: Fluoroquinolone antibiotics are among the most potent second line anti-tubercular drugs these days. The aim of the study was to analyse the frequency and pattern of genetic mutation in pre-extensive (pre-XDR) and extensive drug resistant *Mycobacterium tuberculosis* using second-line line probe assay and to compare drug resistant mutations with different treatment outcomes. **Method:** Sputum, lymph node aspirate and cold accesses from patients with rifampicin resistant TB were subjected to first line and second line Line Probe Assay (Genotype MTBDRsl by Hain Life Science, Germany) to assess additional drug resistance to fluoroquinolones (Levofloxacin & Moxifloxacin). Final treatment outcomes as per National Tuberculosis Elimination Programme was assessed and compared with mutation profile. **Results:** 150 subjects were observed to have mutations associated with resistance to Fluoroquinolones and constituted the final study population. The most frequent mutation observed among Gyr A drug resistance mutation was D94G (Gyr A MUT3C, 44/150, 66%) corresponding to high level resistance to levofloxacin and moxifloxacin. The same mutation was associated with poor treatment outcome as died or treatment failure (OR 2.50, RR1.67, p value 0.043). The most common hetero-resistance mutation pattern observed in Gyr A gene was wild type plus Asp94Gly mutation in 24.6% of isolates. **Conclusions:** GyrA MUT3C hybridization corresponding to single point mutation of Aspartic acid to Glycine at codon 94 was the most common mutation in Gyr A gene locus in *Mycobacterium tuberculosis* and significantly associated with treatment outcome as died compared to those with treatment outcome as cured.

Key words: fluoroquinolones, gene mutation, treatment outcome.

Introduction

Tuberculosis (TB) is a communicable disease with significant morbidity and mortality worldwide. Until the corona virus (COVID-19) pandemic, TB was the leading cause of death from a single infectious agent, ranking above HIV/AIDS.¹ Globally, the estimated number of deaths from TB increased between 2019 and 2021, reversing years of decline between 2005 and 2019.¹ In 2021, there were an estimated 1.4 million deaths among HIV-negative people and 187 000 deaths among HIV-positive people. The estimated incidence of MDR/RR-TB (as per the Global TB Report 2022) in 2021 for India was 119,000 (93,000-145,000). The year 2022 saw an increase of 32% in the number of MDR/RR-TB cases detected under NTEP as compared to 2021.² Globally also the number of people developing MDR/RR TB increased from 437,000 per year in 2020 to 437,000 per year in 2021 [WHO Global TB report 2022].

Rapid and accurate detection of MDR/XDR-TB is of importance as this allows for optimizing therapy and reducing transmission. This is best achieved using molecular tests such as CBNAAT and Line probe assay. MTBDRsl Ver 2.0 assay is one such LPA done for second-line resistance detection in MDR TB/RR TB in India under programmatic conditions. One study found that among FQ-resistant isolates, 18 different types of banding pattern corresponding to mutations were observed.³ The most common banding pattern was MUT3C followed by gyrA MUT1 and gyrA MUT3A respectively.³

Studies relating to different mutations to the treatment outcome in PreXDR TB patients are very limited. Present analysis is an attempt to identify frequency and pattern of these mutations associated with fluoroquinolone resistance and to evaluate different outcomes related to such mutations.

Methods

This retrospective analysis was carried out at Nodal DRTB centre at the Department of Respiratory Medicine and Intermediate Reference Lab. IRL lab is certified by Central TB Division, India, for the testing of molecular and phenotypic drug sensitivity of *Mycobacterium tuberculosis*.

The Genotype MTBDRsl Ver 2.0 assay was performed as per the manufacturer's instructions. In brief, 1 ml of positive liquid culture was centrifuged and the pellet was taken for DNA extraction by GenoLyse kit Hain Lifescience (Nehren, Germany). Before amplification, the kit components AM-A and AM-B were mixed and then the extracted DNA was added and amplified. The hybridization was performed using TwinCubator/GT-Blot and the results were analyzed. The strip contains 27 probes to check internal controls, identification of *M. tuberculosis* complex, and drug targets *gyrA*, *gyrB* for FQ and *rrs*, *eis* for SLID. The missing of wild probe and presence of mutant probe were considered as resistant.

A total of 250 consecutive drug resistant isolates (either MDR TB/ RR TB or isoniazid-resistant) received between 2020 and 2022 were analyzed by routine LPA testing at IRL. Upon completing second line line probe assays of these isolates, FQ resistance was detected in 150 (60%) and mutations in these isolates are studied and compared with final treatment outcome.

Phenotypic DST to moxifloxacin was done in all patients with FQ resistance on SLLPA. 106 / 150 isolates (70.6%) were resistant to moxifloxacin on pDST. 37 / 150 FQ resistant isolates on SLLPA were sensitive on pDST and 7 samples showed no growth on culture.

Treatment regimen – 132/150 patients with FQ resistance were put on all longer oral M/XDR-TB regimen containing BDQ/moxifloxacin/LZD/CFZ/CS. 10/150 patients, who were Rifampicin sensitive and H + FQ resistant were initiated on moxifloxacin/R/Z/E. 8 isolates were resistant to MUT 3B/3C/3D and so were put on combination regimen of BDQ/DLM/CFZ/CS/LZD.

Treatment outcome were assigned as per NTEP guidelines. The data analysis was performed using Epi info version 7.2.1.0. Statistical significance was established at p-value of 0.05. The study was started after obtaining institutional ethical committee approval.

Results

250 rifampicin resistant samples were subjected to second line line probe assay and FQ resistance was detected among 150 (60%). In the study group, 67 % (101/150) were male patients and 33% (49/150) were female patients. Out of total 150 subjects, 130 (86.6%) were from rural area where as 20 (13.3 %) were from urban area. 78 % (116/150) of the subjects were underweight with BMI < 18.5 and 20% (31/150) were in normal range and 2% (3/150) were overweight. 45 % (67/150) subjects were severely underweight. 143 were sputum samples, four were lymph node needle aspirate two cold abscess and one was psoas abscess aspirate. In this study, all subjects were subjected to blood sugar and HIV antibodies estimation. 4 subjects were reactive for HIV antibodies with normal CD4 count and 3 were diabetic controlled on drugs.

Detection of fluoroquinolone resistance

The frequency and mutation profiling of Fluoroquinolone resistance was determined at *Gyr* gene. In total, 149 out of 150 showed resistance for *Gyr A* gene while *Gyr B* gene mutations was present in 1 case with mutation at *Gyr B* WT1. Figure 1 shows all types of mutations in the *Gyr A* gene and the pattern of corresponding wild and mutant bands. Absence of *Gyr A* WT3 band (56%, 84/150) and the development of MUT3C (44%, 66/150) was the most common pattern of mutation. It was followed by

Gyr A MUT1 (18.7%, 28/150) and Gyr A MUT3B (18%, 27/150). Out of 150, 94 samples showed heteroresistance i.e., both wild type and mutant alleles were detected. The most common heteroresistance mutation pattern was a combination of wild type and Asp94Gly (WT plus D94G) in 37 (24.6%) of isolates. This was followed by wild type and Ala90Val (WT + A90V) in 18 (12%) isolates and wild type and Asp94Asn/Tyr in 15 (10%) isolates (WT+D94N/Y). WT plus Asp94His (WT+D94H) was the least common hetero-resistance pattern observed in this study (2%).

Fluoroquinolone resistance mutations and treatment outcome

Table 1 shows treatment outcomes of study subjects. Among 150 subjects, 61 were still on treatment, so outcome could be analysed among 89 patients only. Among these 89 patients, 49 (5%) patients had an outcome as cured and 28 (31.4%) subjects had an outcome as died, 6 (6.7%) had an outcome as treatment failure and 7 (7.8%) were lost to follow up. The most common mutant pattern observed was Gyr A MUT 3C corresponding to D94G mutation in 46.9 % cured patients. Only one wild type mutation in Gyr B gene locus was observed in cured among 150 isolates analysed in this study. The most frequent mutation observed among 28 died subjects was Gyr A MUT3C seen in 13 isolates (46.4 %). The most frequent mutation pattern observed among treatment failure was that of Gyr A MUT 3C in 4 (66%) patients.

Hetero-resistance in Fluoroquinolone and treatment outcome

Table 2 shows comparison of Gyr A hetero-resistance mutations with treatment outcome. Two patients had both MUT1 and MUT3C and both had a poor outcome but statistical significance could not be established (P-value 0.09).

Presence of any Gyr A Mutation was associated with poor treatment outcomes (died and treatment failure) as compared to those without the same mutation [OR:3.71 (CI:1.86-7.42) p=0.0002, Table 3]. The odds ratio for patients with outcome as died or treatment failure among subjects with heteroresistance mutation was calculated to be 0.659 (0.402-1.079) with p-value of 0.106. On comparison of Gyr A MUT 3C mutations with treatment outcome, patients having D94G mutation (Gyr A MUT3C) were observed to have 1.67 times risk of worse treatment outcome (death or treatment failure) as compared to patients without mutation (OR=2.50, CI=1.026-6.088). Gyr A D94G mutation (MUT3C) was significantly associated with treatment outcome as died or treatment failure (53%) as compared to patients with outcome as cured (46.9%) (p-value 0.043).

Discussion

Drug resistance cause prolonged treatment period and increases the risk of poorer treatment outcomes in tuberculosis.⁴ The role of the fluoroquinolone is implied in how extensively drug resistant (XDR) TB is defined, i.e., as TB caused by *Mycobacterium tuberculosis* strains that fulfil the definition of MDR/RR-TB and are also resistant to any fluoroquinolone (levofloxacin or moxifloxacin) and at least one additional Group A drug (either bedaquiline or linezolid or both).⁵ Fluoroquinolones vary in their potency against *Mycobacterium tuberculosis* with third- and fourth-generation agents (e.g., levofloxacin, gatifloxacin, and moxifloxacin) having higher activity than second-generation agents (e.g., ciprofloxacin or ofloxacin).^{6,7}

The global coverage of testing for resistance to fluoroquinolones remains much lower, being 50% in 2021.⁸ The rate of fluoroquinolone resistance prevalence among non-MDR-TB is higher than 0.8% of the global estimate, which may be a warning against the widespread use of fluoroquinolone in the community.⁹ As per WHO report in 2018, 8.5% of all MDRTB patients had additional resistance to

at least one drug.¹⁰ In the present study, FQ resistance among MDR tuberculosis estimates to be 60% (150/250) which is higher than global estimates.

Mutation at codon 94 can be identified by the only commercial molecular test currently available (Genotype MTBDRsl; Hain Lifesciences) by the absence of the WT3 probe, whereas 94Gly mutations can be detected specifically as MUT3C, and 94Ala as MUT3A. The frequency of Gyr A gene mutation (99.3%) was found to be higher than that of Gyr B (0.6%) in the present study which is consistent with the data reported by Kabir et al¹¹ from Pakistan. The finding that development of MUT3C (44%,66/150) was the most common pattern of mutation in FQ resistant tuberculosis isolates is higher than the figure reported by Sethi S et al³ (33%) from Chandigarh, India. It was followed by Gyr A MUT1 (18.7%,28/150) and Gyr A MUT3B (18%,27/150) and the pattern was consistent with that reported by Kabir et al.¹¹

The most common hetero-resistance mutation pattern in Gyr A gene in the present study was a combination of wild type and Asp94Gly (WT plus MUT3C) in 37 (39.3%) of isolates. This was consistent with a study by Rigouts L et al¹² in which the same pattern of hetero resistance was observed in 4 out of 10 (40%) hetero resistance pattern. The least common hetero resistance pattern observed was WT+D94H in 3 (3.1%) isolates. Among 37 isolates having WT+ Asp94Gly mutation, 8 had a treatment outcome as died or failure whereas the rest 18 had an outcome as cured (p -value:0.175). Subjects with a double mutation of Ala90Val and Asp94Gly (Gyr A MUT1 +MUT3C) were all had an outcome as died/failure (p value:0.09). Hetero resistance has been reported to be responsible for limited treatment options and increasing rates of treatment failure in TB patients as reported by Rigouts L et al.¹² In our study, two patients had hetero-resistance at both MUT1 and MUT3C and both had a poor outcome but statistical significance could not be established (p-value 0.09).

The most frequent mutation in Gyr A gene locus among died patients was MUT3C in 46 % patients (13/28). MUT 3C was the most frequent mutation observed among treatment failure case also (66%, 4/6). The present study compared treatment outcome of different mutations and statistically significant difference was noted in subjects with Gyr A MUT3C. Death or failure was significantly higher among those patients. (p-value-0.043). Gyr A hetero-resistance mutations were compared to treatment outcomes but no significant difference could be made (Odd's ratio: 0.659 (CI=0.402-1.079, p value-0.106)

This study observed presence of Gyr A Mutations to be associated with poor treatment outcome (died and treatment failure) as compared to those without the same mutation [OR:3.71 (CI:1.86-7.42) P=0.0002]. This was comparable to a study by Qiao Li et al¹³ from Eastern China who also observed Gyr A gene mutation associated with poor outcome (p=0.002). Statistical analysis of cured and bad treatment outcome (died or treatment failure) was also done and it was observed that the subjects with this mutation had 1.67 relative risk of outcome as died or failure than those without the mutation (OR: 2.500, CI=1.026-6.088, p-value=0.043). Rigouts et al¹² from France reported a higher odds ratio [OR 4.3 (CI 1.4-13.0)] of Asp94Gly mutation predicting a significantly lower proportion of cure compared to the present study. The specific impact of Gyr B mutation on patient outcome could not be analysed because only one sample out of 150 isolates had Gyr B mutation.

Development of mutation leading to resistance to a particular drug does not exclude it from treatment regimen. The A90V (Gyr A MUT 1), S91P (Gyr A MUT2), D94A (Gyr A MUT3A) mutation led to resistance against levofloxacin but a higher generation of FQ (i e Moxifloxacin) can be used at a higher dose. Both Levofloxacin and moxifloxacin are ineffective if D94G mutation is present.

Strengths

The present study compares the FQ resistance mutation pattern among tuberculosis patients in Western India for the first time. Correlation of FQ resistant mutations with treatment outcomes had not been studied earlier. Hetero resistance among fluoroquinolone was also analysed and compared with treatment outcomes, making this study unique.

Limitations

The limitation of this study lies in the absence of sequencing data for confirmation of different drug-resistant related mutations. LPA is less efficient than conventional culture-based DST in detecting resistance in samples that harbour both drug-susceptible and -resistant bacteria (i.e., hetero-resistance). Moreover, there were less extra-pulmonary samples for mutation profile analysis and hence effect of mutations in extra-pulmonary tuberculosis could not be studied in detail. Impact of co-morbid illnesses couldn't be analysed as there were a few cases with such illnesses. This study focussed only on the FQ mutations as data of additional drug resistance was unavailable and therefore their impact on treatment outcome couldn't be studied. Being a hospital based retrospective study, findings cannot be generalized to the community at large, but despite above limitations it gives valuable information regarding the mutation trends of FQ resistant TB cases and associated treatment outcomes.

Conflicts of Interest: None

Acknowledgments: The authors acknowledge the staff of the Intermediary Reference Laboratory (NTEP) for their skillful technical assistance and State TB Cell for financial support in this study.

Abbreviations: BDQ-Bedaquiline, CFZ- Clofazamine CS-Cycloserine, DLM-Delamanid, DRTB- Drug resistant tuberculosis, FQ-Fluoroquinolone, H-isoniazid, LZD-Linezolid, MDR TB-multi drug resistant TB, pDST-phenotypic Drug Sensitivity test, IRL- Intermediate Reference Laboratory, LPA-Line probe assay, NTEP-National Tuberculosis Elimination Program, OR-odds ratio, RRTB-Rifampicin resistant tuberculosis, SLLPA-Second line line probe assay, SLI- second line injectables, WHO- World Health Organisation, XDR- Extensive drug resistant tuberculosis.

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Figure 1: Bar Diagram showing frequency of Gyr A Mutations (n=150)

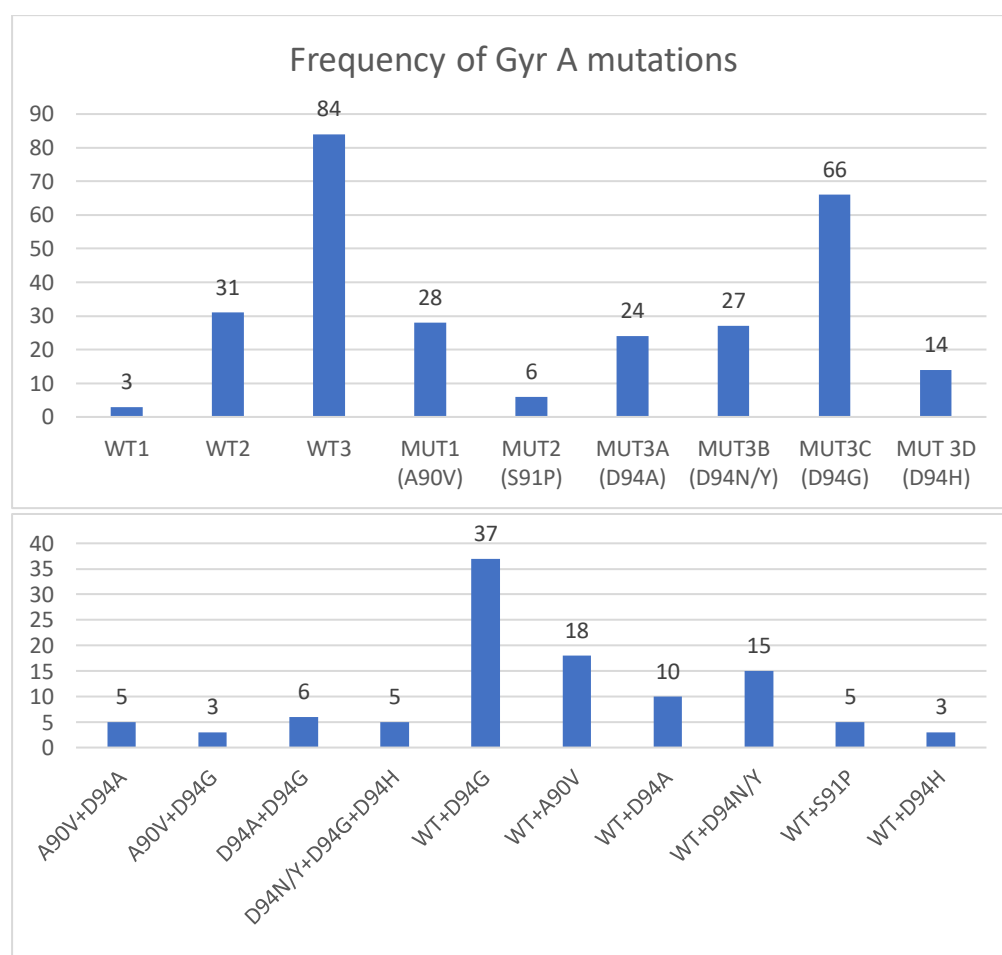


Table 1: Comparison of outcome among FQ resistant isolates (n=89).

Mutation	Cured (N=49) N (%)	Died (N=28) N (%)	Failure (N=6) N (%)	p value
GyrA WT1	2 (4.1)	0	0	0.481
GyrA WT2	16 (32.6)	4 (14.2)	1 (16.7)	0.158
GyrA WT3	29 (59.1)	17 (60.7)	4 (66.7)	0.933
GyrA A90V	11 (22.4)	5 (17.8)	1 (16.7)	0.836
GyrA S91P	3 (6.1)	1 (3.5)	0	0.737
GyrA D94A	4 (8.1)	3 (10.7)	1 (16.7)	0.785
GyrA D94N/Y	9 (18.3)	6 (21.4)	1 (16.7)	0.957
GyrA D94G	23 (46.9)	13 (46.4)	4 (66.7)	0.043
GyrA D94H	4 (8.1)	3 (10.7)	1 (16.7)	0.785
GyrB WT1	1 (2)	0	0	0.697
GyrB N538D	0	0	0	0

Table 2: Frequency of Gyr A multiple mutations and treatment outcome.

Mutation	Number of patients		p value
	Cured	Died/treatment failure	
D94A+ D94G	1	1	0.809
D94N/Y+ D94H	3	3	0.667
D94N/Y+ D94G+ D94H	3	3	0.667
D94N/Y+ D94G	5	3	0.802
A90V+ D94N/Y	1	0	0.395
A90V+ D94G	0	2	0.09
A90V+ D94H	1	0	0.395
WT+ A90V	10	5	0.470
WT+ S91P	3	1	0.488
WT+ D94A	3	2	0.938
WT+ D94N/Y	8	4	0.527
WT+ D94G	18	8	0.175
WT+ D94H	2	1	0.766
WT+ D94N/Y+ D94G	4	0	0.08

Table 2: Comparison of Gyr A mutation patterns with different treatment outcomes.

	Mutation	Cured	Died/failure	OR for outcome Died/failure (95% confidence interval)
Any Gyr A	Present	48	34	3.71 (1.86-7.42)
	Absent	84	16	p=0.0002
Gyr A Wild + mutant (hetero resistance)	Present	35 (65%)	19 (35%)	0.659 (0.402-1.079)
	Absent	14 (47%)	16 (53%)	p=0.106
GyrA MUT3C (D94G)	Present	16 (33%)	20 (55%)	2.500 (1.026-6.088)
	Absent	32 (66%)	16 (44%)	p=0.043
	Z statistic:2.018			
	Relative risk: 1.67			

Reply to the reviewers' comments

Reviewer Number	Original comments of the reviewer	Reply by the author(s)	Changes done on page number and line number
1	The length of paper is too much. The whole manuscript needs to be rewritten with focus on objectives only. Redundant information is not needed. Tables and figures are too many.	As per reviewer`s observation, whole manuscript has been rewritten & length of paper is reduced now.	Page 1-7
2	How cases were selected?	As this was a retrospective study, we enrolled consecutive samples deposited for LPA analysis at respective Intermediate reference lab at our region. This is mentioned at the method section.	Page 2, para 3, line 1
3	What was the treatment provided?	132/150 patients with FQ resistance were put on all Longer oral M/XDR-TB regimen containing BDQ/MOXIFLOXACIN/LZD/CFZ/CS. 10/150 patients, who were Rifampicin sensitive and H + FQ resistant were initiated on MOXIFLOXACIN/R/Z/E. 8 isolates were resistant to MUT 3B/3C/3D and so were put on combination regimen of BDQ/DLM/CFZ/CS/LZD. This is also mentioned in method section.	Page 2, para 5, line 1 ,2
4	Treatment outcome is based on many factors like additional drug resistance. How these factors were	As aptly mentioned, there are many factors which influence treatment outcome. In present study , we focused upon association with fq mutations only as we do not have	Page 4, para 6, line 1,2 Page 5, para 2, line 5

	controlled to decipher association of mutations with outcome.	additional drug resistance data from IRL. This fact is also mentioned as limitation at appropriate place.	
5	Why sequencing was not done to confirm LPA mutations. What was the phenotypic DST result?	As this study was done under NTEP, we do not have data of genome sequencing so it is not performed under guidelines, though it is of importance to confirm LPA result and know about additional drug resistance. This fact is also mentioned in the manuscript. Details of phenotypic DST results are mentioned now.	Page 5, para 2, line1 Page 2, para 4, line 1
6	Comorbidities are important in deciding Tt outcome too. It is a premature analysis.	In our study 4 patients were HIV seropositive with normal CD4 count and 3 were Diabetic, controlled on OHA. There were only few cases having co-morbid illnesses in this study. Although these are important in deciding treatment outcome also, their impact couldn't be analysed due to less number in this study. This has been mentioned in the manuscript.	Page 2, para 6, line 7, 8